

0.5 Polynomials

Definition 0.5.1 — Polynomial

A **polynomial** over a field F in the indeterminate x is an expression of the form:

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = \sum_{k=0}^n a_k x^k$$

where $a_0, a_1, \dots, a_n \in F$ are called the **coefficients** of $p(x)$.

Definition 0.5.2 — Degree

Let $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a nonzero polynomial with $a_n \neq 0$. The **degree** of $p(x)$, denoted $\deg p(x)$, is n . The coefficient a_n is called the **leading coefficient**.

By convention, the zero polynomial $p(x) = 0$ has degree $-\infty$.

Definition 0.5.3 — Polynomial Ring

The set of all polynomials over a field F is denoted $F[x]$:

$$F[x] = \left\{ \sum_{k=0}^n a_k x^k : n \in \mathbb{N}, a_k \in F \right\}$$

This set forms a **ring** under polynomial addition and multiplication. (You can omit what ring means here.)

Example. (Elements of the Polynomial Ring)

The following are elements of $\mathbb{R}[x]$:

$$3x^2 - 2x + 1, \quad x^5 + \pi x^3 - \sqrt{2}, \quad 7, \quad 0$$

Note that constant polynomials (including 0) are also polynomials.

In linear algebra, we often work with polynomials of bounded degree.

Definition 0.5.4 — Polynomials of Degree at Most n

Let F be a field and $n \in \mathbb{N}$. The set of all polynomials over F of degree at most n is denoted:

$$F[x]_{\leq n} = \{p(x) \in F[x] : \deg p(x) \leq n\}$$

Equivalently:

$$F[x]_{\leq n} = \{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 : a_k \in F\}$$

Example. (Polynomials of Degree at Most 2)

The set $\mathbb{R}[x]_{\leq 2}$ consists of all polynomials of degree at most 2:

$$\mathbb{R}[x]_{\leq 2} = \{ax^2 + bx + c : a, b, c \in \mathbb{R}\}$$

Examples include $3x^2 - 2x + 1$, $5x + 7$, and -4 .

Definition 0.5.5 — Polynomials of Degree Exactly n

We use $F[x]_{=n}$ to denote polynomials of degree **exactly** n :

$$F[x]_{=n} = \{p(x) \in F[x] : \deg p(x) = n\}$$

Equivalently:

$$F[x]_{=n} = \{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 : a_n \neq 0, a_k \in F\}$$

Remark. Note the distinction:

- $F[x]_{\leq n}$ contains polynomials of degree $\leq n$ (including the zero polynomial)
- $F[x]_{=n}$ contains polynomials of degree **exactly** n (so $a_n \neq 0$)

Thus $F[x]_{\leq n} \supsetneq F[x]_{=n}$, and in fact $F[x]_{\leq n} = F[x]_{=n} \cup F[x]_{\leq n-1}$.

Example. (Comparing Notations)

The following examples highlight the difference between the notations introduced above:

- $\mathbb{R}[x]_{\leq 2} = \{ax^2 + bx + c : a, b, c \in \mathbb{R}\}$ includes $x^2 + 1$, $3x - 2$, and 5
- $\mathbb{R}[x]_{=2} = \{ax^2 + bx + c : a \neq 0, a, b, c \in \mathbb{R}\}$ includes $x^2 + 1$ but **not** $3x - 2$ or 5